HCI Past Papers

Cambridge CS Tripos | Model Answers + Topic Map

2003 Paper 9, Q1(a) — Mental Models

Cognitive Processes

Topic: Mental Models | Designer vs User Model

Q: Sign says 'Caution Site Entrance' near Gates building. What does this show about mental models?

Key Point: The designer's mental model ≠ the user's mental model.

- Designer assumed: People approaching know it's a building entrance → threat of vehicles.
- User (HCI lecturer) interpreted: A generic caution sign, not specific enough.
- Mental model = user's internal understanding of how a system/environment works.
- Mismatch = usability problem. Good design aligns the designer's model with the user's model.

Exam tip: Always say WHAT the designer assumed + WHAT the user understood + WHY they differ.

2009 Paper 7, Q10(b) — User Behaviour Analysis

Cognitive Processes

Topic: Cognitive Strategies | Mental Models | User Types

Q: Two patterns found in tax return logs — (1) trial & error, (2) long pauses + help seeking. Explain them.

Pattern 1: Trial & Error (exploratory users)

- Users treat the system as explorable — they test values to understand outcomes.
- Mental model: system is a sandbox → action → observe result → revise.
- Fix: Better feedback after each input (show consequences before committing).

Pattern 2: Long pauses + Help seeking (confused users)

- Users don't know what to enter → consult help in alternation with data entry.
- Mental model is incomplete — they need external scaffolding.
- Fix: Inline contextual help, tooltips, examples embedded on the form itself.

Exam tip: Name the cognitive strategy (exploratory vs. help-dependent), then suggest fix.

2008 Paper 7, Q12 — Design Process for Older Users

Design Principles + Prototyping

Topic: User-Centred Design | Prototyping | Usability Evaluation

Q: Describe 5 UI design techniques at different stages of the design cycle for older users (MP3/radio players).

Stage 1 — Requirements: Contextual Inquiry — observe older users with current devices in their homes. Reveals real pain points (small fonts, complex menus).

Stage 2 — Conceptual Design: Persona creation — build personas of older users (low tech literacy, large hands, vision issues). Guides design decisions.

Stage 3 — Prototyping: Low-fi paper prototype — sketch large buttons, simple 2-level menu. Cheap, fast to test.

Stage 4 — Evaluation: Think-aloud usability testing — older users perform tasks while narrating thoughts. Reveals confusion points.

Stage 5 — Iteration: Heuristic evaluation — experts check against Nielsen's 10 heuristics (esp. error prevention, visibility of system status).

Exam tip: Link EACH technique to the design stage + explain WHY it suits older users.

2005 Paper 9, Q1 — Interaction Design + Prototyping

Design Principles + Prototyping

Topic: GOMS/KLM | Prototyping | Task Analysis

(a) Estimate speed increase from adding 'paste special' to right-click menu.

Use **KLM (Keystroke Level Model)**: count operators (K=keystroke, M=mental op, P=pointing) for old vs new method.

- Old: Right-click → navigate to Edit menu → Paste → open Paste Special dialog → click → OK.
- New: Right-click → Paste Special directly visible → click. Fewer steps = measurably faster.
- Calculate total time: T = sum of all operator times. Compare old vs new.

(b) Confirm actual speed increase after prototype.

Build a working prototype → run **controlled experiment**:

- Participants: sample of banking users.
- Task: copy value from spreadsheet, paste into letter — repeat N times.
- Measure: task completion time (stopwatch or logging).
- Analysis: t-test comparing mean time old vs new. $p < 0.05$ = significant improvement.

(c) What factors assess the full 'word processor with calc functions' alternative?

- Task coverage: Does it support ALL spreadsheet tasks users currently do?
- Learnability: Can banking users learn the new calc functions without training?
- Error rate: More likely to make errors in unfamiliar environment?
- User satisfaction: Qualitative interviews / Likert scale.
- Workflow disruption: How much existing work practice changes.

Exam tip: (a) = analytic method. (b) = empirical summative. (c) = multi-factor UX assessment.

2003 Paper 9, Q1(b-c) — Heuristic Eval + Cognitive Walkthrough

Evaluation Methods

Topic: Heuristic Evaluation | Cognitive Walkthrough | Kiosk Interface

Q: Apply heuristic evaluation AND cognitive walkthrough to the lecture control interface. Find 2 problems each.

Heuristic Evaluation (Nielsen's 10):

- Problem 1 — Visibility of system status: No feedback shown when a system preset is selected (Screen B). User doesn't know if action was registered.
- Problem 2 — Consistency & standards: Screen A has fixed function keys around the screen but labelled differently from main menu options — violates consistency.

Cognitive Walkthrough:

For each step, ask: (1) Will user know what to do? (2) Will they notice the correct control? (3) Will they understand feedback?

- Problem 1 — Screen A: 'QUICK keys' label is ambiguous. User's goal is to control audio/lighting but the label doesn't map to goal → user won't notice correct action.
- Problem 2 — Screen B: After selecting a preset, no confirmation appears. User can't tell if selection worked → no feedback loop closes the action.

(c) Sketch alternative design for one screen + Cognitive Dimensions trade-off.

Alternative for Screen A: Replace fixed-position quick keys with a grid menu showing icons + text labels for Lights, Music, Projection.

- CD Trade-off: Improves Visibility (easier to find options) BUT reduces Terseness (takes more space, more steps for expert users).

Exam tip: Cognitive Walkthrough = user's perspective, step-by-step. Heuristic = expert judges against principles.

2016 Paper 7, Q9(a-d) — Formative vs Summative Research

Evaluation Methods

Topic: Formative | Summative | Empirical | Analytic | Wearable Device

(a) Difference between formative and summative HCI research.

- **Formative:** Done DURING development. Goal = find problems to fix. E.g. usability testing a prototype.
- **Summative:** Done AFTER development. Goal = measure final performance. E.g. compare two finished systems.

(b) Two formative empirical methods — one qualitative, one quantitative.

Qualitative formative: Semi-structured interviews with potential wearable device users.

- Steps: Recruit 8-10 users → interview about social situations they struggle with → thematic analysis.
- Good data qualities: Rich, contextual, captures nuance of social obligation types.

Quantitative formative: Diary study — participants log social challenges over 2 weeks.

- Steps: Provide structured diary template → count frequency of situation types.
- Good data qualities: Objective counts, representative sample of real situations.

(c) Two summative empirical methods — one qualitative, one quantitative.

Qualitative summative: Observational study — users wear device in real social situations.

- Collect: Video + notes of interactions. Analyse: thematic coding of cue-response behaviour.

Quantitative summative: Controlled experiment — measure response accuracy to device cues.

- Measure: % correct social responses, reaction time. Analyse: descriptive stats + t-test.

(d) Two analytic research methods — one qualitative, one quantitative.

Qualitative analytic: Heuristic evaluation — experts assess wearable UI against Nielsen's 10 heuristics.

- Expresses options in qualitative terms: identifies specific usability violations + severity ratings.

Quantitative analytic: KLM (Keystroke Level Model) — model time to configure device via web interface.

- Expresses options in quantitative terms: predicted task time in seconds for each configuration flow.

Exam tip: Empirical = involves users. Analytic = no users needed (experts or models). Always state which is which.

2017 Paper 7, Q8(b-c) — Empirical Research Methods

Evaluation Methods

Topic: Formative | Summative | ML vs Programming Language | Empirical Research Design

(b) Propose a FORMATIVE empirical method to compare ML vs programming language approach for home lighting.

Method: Field study / in-home observation

- Step 1: Recruit 10 households (potential customers, varied tech literacy).
- Step 2: Give each household both systems for 2 weeks each (counterbalanced order).
- Step 3: Observe how they naturally interact — do they create lighting schemes? How often do they use it?
- Data collected: Usage logs (frequency, schemes created), video of interaction, weekly interviews.
- Analysis: Thematic analysis of interviews + compare usage frequency between systems.
- Recommendations: Which approach better supports natural home life? What specific problems need fixing?

(c) Propose a SUMMATIVE empirical method to compare speed of defining a lighting scheme quickly (e.g. guests arriving).

Method: Controlled experiment (lab-based task timing)

- Step 1: Recruit 20 participants, no prior experience with either system.
- Step 2: Give standardised task: 'Guests arriving in 2 mins — set up party lighting for living room.'
- Step 3: Each participant does task on BOTH systems (within-subjects design, counterbalanced).
- Data collected: Task completion time (seconds), success/failure, number of errors.
- Analysis: Paired t-test on completion times. If $p < 0.05$, one system is significantly faster.
- Recommendations: Recommend faster system for time-critical use; flag which system fails more.

Key distinction: Formative = open, real-world, finds problems. Summative = controlled, measures final performance.

2006 Paper 9, Q1 — Heuristic Evaluation (Book Website)

Evaluation Methods

Topic: **Heuristic Evaluation | Nielsen's Heuristics | Usability**

(a) Has the manager misunderstood heuristic evaluation?

Yes, partially. The manager has confused heuristics with design rules/specifications.

- Real heuristic eval: Experts judge a system against established heuristics (e.g. Nielsen's 10).
- Manager's heuristics 1 & 3 are actually quantifiable design rules, not heuristic principles.
- Heuristic 2 ('good match between buttons and goals') is more legitimately a heuristic — it matches Nielsen's 'Match between system and real world'.

(b) Comment on each of the 3 proposed heuristics.

Heuristic 1: '5-9 navigation options per page'

- Theoretical basis: Miller's Law (7±2 chunks in working memory) — but this applies to memory, not necessarily navigation menus on screen.
- Additional steps: Evaluate whether options are meaningful and task-relevant, not just count them.
- Design impact: Could artificially limit or inflate menus to hit the number rather than serve user goals.

Heuristic 2: 'Good match between buttons and users' goals'

- Theoretical basis: Strongly grounded — matches Nielsen's H2 (match between system and real world).
- Additional steps: Task analysis needed to identify real user goals first, then evaluate match.
- Design impact: High — directly targets whether navigation labels make sense to users.

Heuristic 3: 'Easy to change plans'

- Theoretical basis: Relates to Nielsen's H3 (user control & freedom) — undo/redo, back navigation.
- Additional steps: Test specific scenarios — can user edit basket, change delivery, cancel order?
- Design impact: Ensures error recovery is possible; important for e-commerce where users change minds.

Exam tip: For each heuristic: state if valid + which Nielsen heuristic it maps to + what extra steps needed + design impact.

2007 Paper 8, Q14 — Usability for Different User Groups

Evaluation Methods

Topic: **Heuristic Evaluation | CDN | User Groups | Usability Requirements**

(a) How would usability requirements differ for Staff vs Graduate Applicants?

Staff (tutorial office workers):

- Frequent, expert users → efficiency matters most. Need keyboard shortcuts, batch operations.
- Error recovery critical — staff process many records, mistakes must be undoable.
- Learnability less important (one-time learning), memorability matters more.

Graduate Applicants:

- Infrequent, novice users → learnability is the top priority.
- Clear instructions, no assumed knowledge, strong error messages (they can't ring IT support).
- Satisfaction matters — poor UX = lost applicants = reputational damage.

(b) Which usability analysis methods for each group?

For Staff: Contextual inquiry (observe in their office environment).

- Why: Reveals real workflow, interruptions, multi-tasking context.
- Analysis: Task flow diagrams, identify bottlenecks in repetitive actions.

For Applicants: Think-aloud usability testing with prototype.

- Why: Reveals where first-time users get confused without external support.
- Analysis: Code think-aloud transcripts for confusion points, measure task success rate.

(c) Three interaction features addressing both groups (using Heuristic Eval / CDN).

- **Feature 1 — Dual interface mode:** Simplified view for applicants, power view for staff. Addresses learnability vs efficiency trade-off.
- **Feature 2 — Inline error messages with guidance:** Helps applicants recover from errors (Nielsen H9). Staff get brief error codes.
- **Feature 3 — Progressive disclosure:** Show basic fields first, advanced options hidden. CDN: reduces Viscosity for novices, keeps options accessible for experts.

Exam tip: Name the principle (Nielsen heuristic or CD) — don't just describe the feature.

2015 Paper 7, Q8 — Cognitive Dimensions (Website Analysis)

Cognitive Dimensions

Topic: Cognitive Dimensions of Notations | Visual Language | Website Design

(a) Name a website and describe its information structure.

Example: **YouTube**. Information structure = hierarchical + navigational. Users interact with: subscriptions (list), playlists (collections), video pages (single item), search results (flat list). The structure has multiple entry points to the same content.

(b) Two aspects of visual language and their meaning.

Aspect 1 — Marks (icons): Thumbs-up icon = 'Like'. The visual mark (upward thumb) maps directly to cultural gesture of approval → strong perceptual correspondence. User knows the meaning without reading text.

Aspect 2 — Regions (layout zones): Left sidebar = navigation/subscriptions. Main area = content feed. Region boundary signals functional separation → user builds spatial mental model of where things live.

(c) A typical activity + two Cognitive Dimensions that affect user experience.

Activity: User adds a video to a playlist while browsing.

CD 1 — Visibility: The 'Save' option is hidden under a '...' menu → poor visibility. User has to hunt for it. If it were visible as a button, task would be faster and more discoverable.

CD 2 — Viscosity: To create a new playlist and add to it simultaneously requires multiple steps (click save → add to playlist → create new → name it → save). High viscosity → friction for users who want to organise while watching.

(d) Propose a visual design modification affecting one CD + trade-offs.

Modification: Add persistent 'Save to playlist' button below each video.

- Effect on CD: Improves Visibility — action is always available without hunting.
- Trade-off: Reduces Terseness — interface becomes more cluttered. Less clean for users who never use playlists.

(e) How to investigate the predicted effects.

Method: A/B usability test — Group A uses original UI, Group B uses modified UI.

- Task: 'Save this video to a playlist in under 30 seconds.'
- Measure: Task completion time, error rate, user satisfaction (post-task Likert scale).
- Analysis: Compare means using t-test. Observe whether modification improves speed without increasing errors.

(f) Classify the method in (e).

Qualitative/Quantitative: Both — task time is quantitative; satisfaction scale is quantitative; but observations during testing are qualitative.

Empirical/Analytic: Empirical — involves real users performing tasks.

Summative/Formative: Summative — comparing two finished designs to see which performs better.

Exam tip: In classification questions, state ALL three dimensions.

2013 Paper 7, Q8(b) — Cognitive Dimensions (Bicycle HUD)

Cognitive Dimensions

Topic: Cognitive Dimensions | Wearable/AR | Tangible UI | Audible Feedback

(b) Three Cognitive Dimensions for modifying appointment schedules while riding.

Context: User wants to edit appointments while cycling — dangerous if interaction is complex.

CD 1 — Viscosity (resistance to change):

- Problem: High viscosity = many steps to reschedule appointment while riding = dangerous distraction.
- Implication: Minimise steps. Use voice commands to reduce viscosity (say 'move 3pm to 4pm').

CD 2 — Visibility:

- Problem: HUD must show appointment info without occluding road view.
- Implication: Use peripheral display or audio cues. Only show most urgent appointment, not full calendar.

CD 3 — Error-proneness:

- Problem: Gesture/gaze input while moving increases accidental triggers.
- Implication: Require confirmation gesture before committing any schedule change. Undo must be 1-step.

(c) Tangible UI vs AR Interface for bicycle — sensor processing.

(i) Tangible UI: Physical controls on handlebars (buttons/dials).

- Sensor processing: Physical button press → digital signal → lookup appointment → display on HUD.
- Advantage: No recognition needed; direct physical mapping.

(ii) Augmented Reality: Overlay appointment info onto real-world view via HUD.

- Sensor processing: Camera captures road scene → computer vision detects road/landmarks → overlay calendar info at correct position relative to scene.
- Also needs: GPS for location-aware reminders, IMU (accelerometer/gyroscope) for head tracking.

Exam tip: For tangible = physical sensors. For AR = camera + vision + tracking. Always name the sensors.

2010 Paper 7, Q10(a) — CDN for Mobile Programming

Cognitive Dimensions

Topic: Cognitive Dimensions | Mobile/Handheld | Programming Environment

(a) Four CDN usability issues for programming on a mobile phone.

CD 1 — Visibility:

- Small screen limits how much code is visible at once.
- User can't see variable defined 5 lines above while editing current line → must scroll constantly.

CD 2 — Viscosity:

- Editing code on mobile is slow — virtual keyboard, tiny tap targets, frequent typos.
- Renaming a variable = high viscosity (many instances to change, hard to select text precisely).

CD 3 — Error-proneness:

- Touch input on small screen = high chance of tapping wrong character or bracket.
- Syntax errors common; need strong real-time error highlighting to catch early.

CD 4 — Consistency:

- Mobile keyboard auto-corrects code keywords (e.g. 'def' → 'the') → breaks consistency between what user types and what appears.
- Design fix: Disable autocorrect in code input fields; provide language-specific keyboard.

Exam tip: Always give a SPECIFIC example of the issue in context — don't just define the CD.

Topic Coverage Map

Topic	Papers Covered	Key Concepts
Cognitive Processes	2003 P9Q1(a), 2009 P7Q10(b)	Mental models, user strategies, trial-error vs help-seeking
Design Principles + Prototyping	2008 P7Q12, 2005 P9Q1, 2003 P9Q1(b-c)	User-centred design stages, KLM, paper prototyping, task analysis
Evaluation Methods	2016 P7Q9, 2017 P7Q8, 2006 P9Q1, 2007 P8Q14	Formative/summative, empirical/analytic, qualitative/quantitative, heuristic eval, cognitive walkthrough
Cognitive Dimensions of Notations	2015 P7Q8, 2013 P7Q8(b), 2010 P7Q10(a)	Visibility, viscosity, error-proneness, consistency, progressive evaluation
Wearable & Mobile Devices	2016 P7Q9, 2013 P7Q8, 2010 P7Q10(a)	Wearable UI design, AR/tangible interfaces, mobile constraints, sensor processing

Quick Recall: Key Distinctions

Term	Means	Example
Formative	During development, find problems	Interview users while prototyping
Summative	After development, measure performance	Timed task test on finished system
Empirical	Involves real users	Think-aloud study, A/B test
Analytic	No users — experts or models	Heuristic eval, KLM, CW
Qualitative	Non-numerical, descriptive	Interview themes, observations
Quantitative	Numerical, measurable	Task time, error count, Likert scores
Heuristic Eval	Experts check against known principles	Nielsen's 10 heuristics
Cognitive Walkthrough	Simulate user's step-by-step cognition	Will user know what to do here?
KLM	Predict task time from operator counts	K+P+M time = total predicted time